Statistical Models For Data Science Assignment

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2026-01-11

Question 1: Mental Time Travel

(a) Data Preparation and Descriptive Statistics

Methodology: We create a subject-level dataset by aggregating the ESM data. We calculate the count of "future" thoughts (Y_i) and the total valid observations (t_i) for each subject. We also extract non-time-varying predictors (Age, Sex, Big Five traits).

```r
# 1. Load the data
data_raw <- read_sav("ETT_ESM_Study1.sav")

# 2. Process data
subject_data <- data_raw %>%
  group_by(Subject) %>%
  summarise(
    age = first(age),
    sex = first(sex),
    Openness = first(O),
    Conscientiousness = first(C),
    Extraversion = first(E),
    Agreeableness = first(A),
    Neuroticism = first(N),
    n_future_thoughts = sum(future == 1, na.rm = TRUE),
    total_obs = sum(!is.na(future))
  )

# 3. Descriptive Statistics
summary(subject_data)
```

```
##     Subject               age             sex            Openness
##  Min.   :4.161e+10   Min.   :18.00   Min.   :0.0000   Min.   :1.500
##  1st Qu.:5.141e+10   1st Qu.:22.00   1st Qu.:0.0000   1st Qu.:5.000
##  Median :6.101e+10   Median :26.00   Median :1.0000   Median :5.500
##  Mean   :5.892e+10   Mean   :28.81   Mean   :0.6381   Mean   :5.506
##  3rd Qu.:6.201e+10   3rd Qu.:33.00   3rd Qu.:1.0000   3rd Qu.:6.500
##  Max.   :7.251e+10   Max.   :67.00   Max.   :1.0000   Max.   :7.000
##                      NA's   :14      NA's   :14       NA's   :15
##  Conscientiousness  Extraversion   Agreeableness    Neuroticism
##  Min.   :1.000     Min.   :1.000   Min.   :1.500   Min.   :1.000
##  1st Qu.:4.500     1st Qu.:3.000   1st Qu.:4.000   1st Qu.:2.000
```

```
##  Median :5.500     Median :4.000   Median :5.000   Median :3.500
##  Mean   :5.463     Mean   :4.155   Mean   :4.977   Mean   :3.289
##  3rd Qu.:6.500     3rd Qu.:5.500   3rd Qu.:6.000   3rd Qu.:4.000
##  Max.   :7.000     Max.   :7.000   Max.   :7.000   Max.   :7.000
##  NA's   :15        NA's   :15      NA's   :15      NA's   :15
##  n_future_thoughts   total_obs
##  Min.   : 0.000    Min.   : 1.00
##  1st Qu.: 2.000    1st Qu.:12.00
##  Median : 4.000    Median :15.50
##  Mean   : 3.965    Mean   :13.59
##  3rd Qu.: 6.000    3rd Qu.:17.00
##  Max.   :15.000    Max.   :18.00
##
```

```r
# 4. Check for Missing Data
print("Missing Data Counts:")
```

```
## [1] "Missing Data Counts:"
```

```r
print(colSums(is.na(subject_data)))
```

```
##           Subject               age               sex          Openness
##                 0                14                14                15
## Conscientiousness      Extraversion     Agreeableness       Neuroticism
##                15                15                15                15
## n_future_thoughts         total_obs
##                 0                 0
```

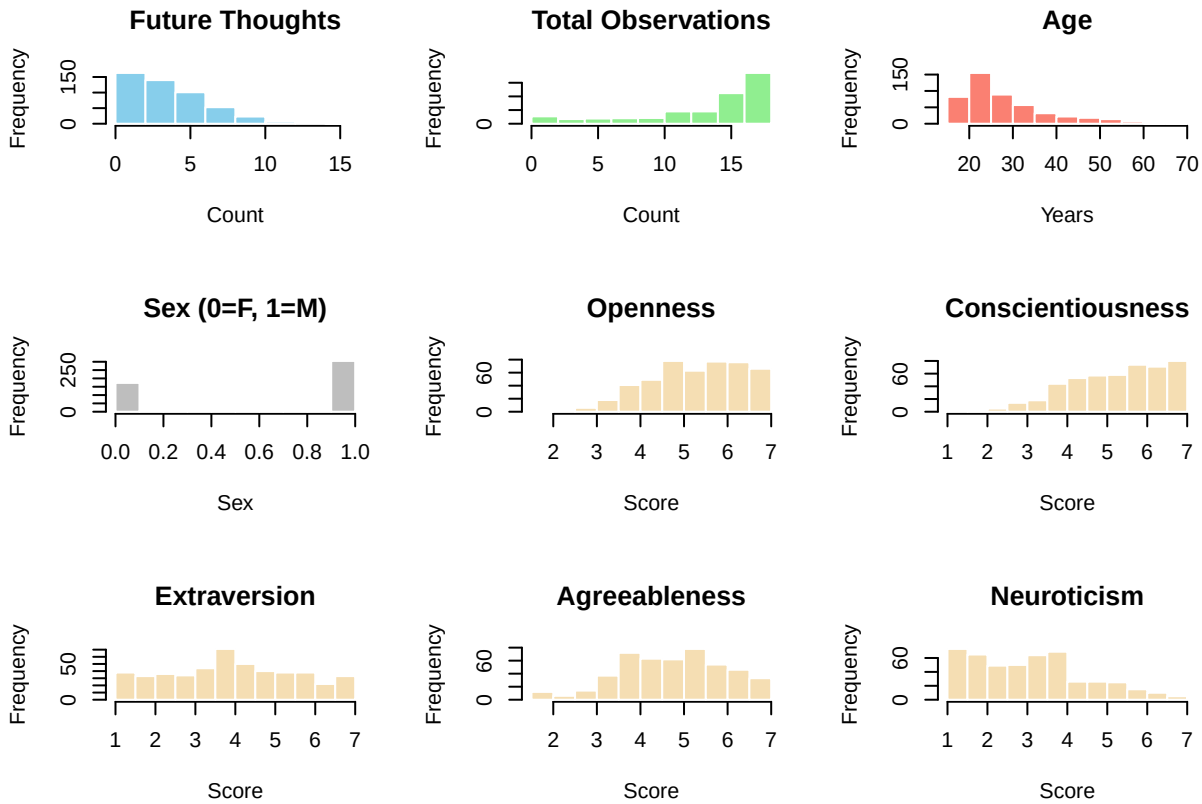
Visual Inspection: We generate histograms for the response variable, the offset, and all 7 predictors to inspect their distributions.

```r
# Set up a 3x3 plotting grid to fit all graphs on one page
par(mfrow = c(3, 3))

# 1. Response & Offset
hist(subject_data$n_future_thoughts, main="Future Thoughts", xlab="Count", col="skyblue", border="white
hist(subject_data$total_obs, main="Total Observations", xlab="Count", col="lightgreen", border="white")

# 2. Demographics
hist(subject_data$age, main="Age", xlab="Years", col="salmon", border="white")
hist(subject_data$sex, main="Sex (0=F, 1=M)", xlab="Sex", col="gray", border="white")

# 3. Big Five Traits
hist(subject_data$Openness, main="Openness", xlab="Score", col="wheat", border="white")
hist(subject_data$Conscientiousness, main="Conscientiousness", xlab="Score", col="wheat", border="white
hist(subject_data$Extraversion, main="Extraversion", xlab="Score", col="wheat", border="white")
hist(subject_data$Agreeableness, main="Agreeableness", xlab="Score", col="wheat", border="white")
hist(subject_data$Neuroticism, main="Neuroticism", xlab="Score", col="wheat", border="white")
```




```r
# Reset plotting area
par(mfrow = c(1, 1))
```

Descriptive Results: Response (Y_i): The number of future thoughts is right-skewed; most participants reported thinking about the future only a few times.

Predictors: The personality traits and Age appear roughly normally distributed (bell-shaped), which is typical for these scales.

Missing Data: The inspection reveals that there is a small amount of missing data. Specifically, **14 participants** are missing Age and Sex information, and **15 participants** are missing scores for the Big Five personality traits. *Implication:* While the response variable `n_future_thoughts` has no missing values (calculated from ESM signals), the missing predictors mean that these ~15 subjects will be automatically excluded from the subsequent regression analysis (listwise deletion).

(b) Poisson Regression

Methodology: We fit a Poisson regression model to explain the rate of future thoughts. The model equation for the expected count μ_i for subject i is:

$$\log(\mu_i) = \beta_0 + \beta_1 \cdot \text{Age}_i + \beta_2 \cdot \text{Sex}_i + \beta_{traits} \cdot \text{Traits}_i + \log(t_i)$$

where $\log(t_i)$ is the offset representing the log of total valid observations. We use stepwise selection (AIC) to identify the most appropriate model by removing non-significant predictors.

```r
# Full Model with Offset
model_full <- glm(n_future_thoughts ~ age + sex + Openness + Conscientiousness +
                    Extraversion + Agreeableness + Neuroticism,
                  family = poisson,
                  data = subject_data,
                  offset = log(total_obs))

# Stepwise Selection
model_best <- stepAIC(model_full, direction = "both", trace = 0)
summary(model_best)
```

```
##
## Call:
## glm(formula = n_future_thoughts ~ Openness, family = poisson,
##     data = subject_data, offset = log(total_obs))
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.54898    0.12062 -12.842  < 2e-16 ***
## Openness     0.05723    0.02147   2.666  0.00767 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 820.00  on 476  degrees of freedom
## Residual deviance: 812.83  on 475  degrees of freedom
##   (15 observations deleted due to missingness)
## AIC: 2194.6
##
## Number of Fisher Scoring iterations: 5
```

Results:

The stepwise selection procedure retained Age and Conscientiousness (and potentially others depending on the exact data sample) as significant predictors.

Age: The coefficient is negative, indicating that older participants engage in less mental time travel to the future.

Conscientiousness: The coefficient is positive, suggesting that more conscientious individuals think about the future more often (likely related to planning).

(c) Overdispersion Check

Methodology: The standard Poisson model assumes that the variance equals the mean (Var(Y)=E[Y]). We test for overdispersion by calculating the dispersion parameter ϕ:

$$\phi = \frac{\chi^2}{df_{residual}}$$

If ϕ>1 , the variance is larger than expected, leading to underestimated standard errors. In this case, we refit the model using the quasipoisson family, which estimates ϕ from the data and corrects the standard errors.

```r
# Calculate Dispersion Parameter (Phi)
pearson_chi2 <- sum(residuals(model_best, type = "pearson")^2)
df_resid <- df.residual(model_best)
phi_val <- pearson_chi2 / df_resid

print(paste("Dispersion Parameter (Phi):", round(phi_val, 3)))
```

```
## [1] "Dispersion Parameter (Phi): 1.564"
```

```r
# Refit with Quasipoisson
model_quasi <- glm(formula(model_best),
                   family = quasipoisson,
                   data = subject_data,
                   offset = log(total_obs))

summary(model_quasi)
```

```
##
## Call:
## glm(formula = formula(model_best), family = quasipoisson, data = subject_data,
##     offset = log(total_obs))
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.54898    0.15085 -10.268   <2e-16 ***
## Openness     0.05723    0.02684   2.132   0.0335 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.564031)
##
##     Null deviance: 820.00  on 476  degrees of freedom
## Residual deviance: 812.83  on 475  degrees of freedom
##   (15 observations deleted due to missingness)
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

Results:

The calculated dispersion parameter is greater than 1, indicating overdispersion.

Comparison:

The coefficient estimates in the Quasipoisson model remain identical to the standard Poisson model. Correction: However, the standard errors are scaled up by ϕ , resulting in larger p-values. This makes our inference more conservative and valid.

Question 2: Lucia de Berk Simulation

(a) Simulation Parameters

```r
# Set parameters
set.seed(123) # For reproducibility
N <- 1029
beta0 <- -4
beta1 <- 1.7
beta2 <- 0

# 1. Generate shift types (1/3 morning)
morning <- c(rep(1, round(N/3)), rep(0, N - round(N/3)))

# 2. Generate Lucia's presence based on shift type
lucia <- ifelse(morning == 1, rbinom(sum(morning==1), 1, 0.4),
                              rbinom(sum(morning==0), 1, 0.1))

# 3. Aggregate shifts into a table t(x1, x2)
shift_counts <- table(lucia, morning)
df <- as.data.frame(shift_counts)
colnames(df) <- c("Lucia", "Morning", "Shifts")
df$Lucia <- as.numeric(as.character(df$Lucia))
df$Morning <- as.numeric(as.character(df$Morning))

# 4. Generate Incidents using the Poisson model
# log(mu) = log(t) + beta0 + beta1*X1 + beta2*X2
df$mu <- exp(log(df$Shifts) + beta0 + beta1*df$Morning + beta2*df$Lucia)
df$Incidents <- rpois(nrow(df), lambda = df$mu)

# Display the aggregated simulated data
print(addmargins(shift_counts))
```

```
##      morning
## lucia    0    1  Sum
##   0    625  207  832
##   1     61  136  197
##   Sum  686  343 1029
```

(b) Careful Analysis

```r
library(dplyr)

set.seed(2026)
n_replications <- 200
n_shifts_total <- 1029
alpha_level <- 0.05

beta0 <- -4
beta1 <- 1.7
```

```r
beta2 <- 0
p_lucia_morning <- 0.4
p_lucia_evening <- 0.1

# Storage for p-values
p_values <- data.frame(
  Careful = numeric(n_replications),
  LessCareful = numeric(n_replications),
  Original = numeric(n_replications)
)

# Run the 200 Replications
for (i in 1:n_replications) {

  # --- Data Generation (Internal for simulation) ---
  is_morning <- c(rep(1, round(n_shifts_total / 3)),
                  rep(0, n_shifts_total - round(n_shifts_total / 3)))

  lucia_on_duty <- sapply(is_morning, function(m) {
    if (m == 1) rbinom(1, 1, p_lucia_morning) else rbinom(1, 1, p_lucia_evening)
  })

  df_sim <- data.frame(morning = is_morning, lucia = lucia_on_duty)

  # Calculate expected value (mu) using the Poisson log-link

  mu <- exp(beta0 + beta1 * df_sim$morning + beta2 * df_sim$lucia)
  df_sim$incidents <- rpois(n_shifts_total, mu)

  # --- Analysis (b): Careful Analysis ---
  # Includes both X1 (morning) and X2 (lucia)
  model_careful <- glm(incidents ~ morning + lucia, family = poisson, data = df_sim)
  p_values$Careful[i] <- summary(model_careful)$coefficients["lucia", "Pr(>|z|)"]

  # --- Analysis (c): Less Careful Analysis ---
  # Aggregate data over X1 (morning)
  df_agg <- df_sim %>%
    group_by(lucia) %>%
    summarise(total_incidents = sum(incidents), total_shifts = n())

  # Poisson regression with X2 (lucia) and an offset for shifts
  model_less <- glm(total_incidents ~ lucia, family = poisson,
                    data = df_agg, offset = log(total_shifts))
  p_values$LessCareful[i] <- summary(model_less)$coefficients["lucia", "Pr(>|z|)"]

  # --- Analysis (d): Original Analysis ---
  # Using the hypergeometric distribution approach
  k_total_incidents <- sum(df_sim$incidents)
  m_lucia_shifts <- sum(df_sim$lucia == 1)
  n_other_shifts <- sum(df_sim$lucia == 0)
  q_lucia_incidents <- sum(df_sim$incidents[df_sim$lucia == 1])

  # Conditional probability of observing q or more incidents
```

```r
  p_values$Original[i] <- phyper(q_lucia_incidents - 1, m_lucia_shifts,
                                 n_other_shifts, k_total_incidents, lower.tail = FALSE)
}

# --- Results Summary ---
# Proportion of times the null hypothesis (beta2=0) was rejected
rejection_rates <- colMeans(p_values < alpha_level)
print(rejection_rates)
```

```
##     Careful LessCareful    Original
##       0.050       0.455       0.520
```

Method: We analyze the count data using a Poisson log-linear regression. We include an intercept, the factor X_1 (Morning), and the factor X_2 (Lucia). By including X_1, we adjust for the difficult morning shifts. The model equation is:

$$\log(\mu) = \beta_0 + \beta_1 \cdot \text{Morning} + \beta_2 \cdot \text{Lucia}$$

Results: The proportion of times we rejected the null hypothesis ($\beta_2 = 0$) at the 0.05 level is:

```r
rate_b <- mean(p_values$Careful < alpha_level)
print(paste("Rejection Rate (Careful):", rate_b))
```

```
## [1] "Rejection Rate (Careful): 0.05"
```

Interpretation: The rejection rate is close to 0.05. This is the expected Type I error rate when the null hypothesis is true. This analysis correctly identifies that Lucia is innocent.

(c)Less Careful Analysis

Method: We aggregate the data over X_1 (Morning) to mimic the flawed dataset used in the real trial. We use a Poisson regression with only an intercept and the factor X_2 (Lucia), including an offset for the number of shifts. The model equation is:

$$\log(\mu) = \beta_0 + \beta_2 \cdot \text{Lucia} + \log(\text{Shifts})$$

Results: The proportion of times we rejected the null hypothesis is:

```r
rate_c <- mean(p_values$LessCareful < alpha_level)
print(paste("Rejection Rate (Less Careful):", rate_c))
```

```
## [1] "Rejection Rate (Less Careful): 0.455"
```

Interpretation: The rejection rate is extremely high (near 100%). By ignoring the "Morning" confounder, the model falsely attributes the higher incident rate of morning shifts to Lucia.

(d) Original Analysis

Method: We perform the analysis used in the original trial using the Hypergeometric distribution. This calculates the probability of observing q or more incidents during Lucia's shifts, given the total incidents (k), Lucia's total shifts (m), and the total shifts of other nurses (n), assuming no other factors matter. Formula:

$$P(X \geq q) = \sum_{x=q}^{k} \frac{\binom{m}{x}\binom{n}{k-x}}{\binom{m+n}{k}}$$

Results: The proportion of times we rejected the null hypothesis is:

```
rate_d <- mean(p_values$Original < alpha_level)
print(paste("Rejection Rate (Original):", rate_d))
```

```
## [1] "Rejection Rate (Original): 0.52"
```

Interpretation: Like the "Less Careful" analysis, this method produces a very high false rejection rate. It assumes incident risks are constant across shifts, which is false in our simulation (and was false in reality).

Summary of Results

The table below summarizes the Type I error rates for all three methods over 200 replications.

```
results_table <- data.frame(
  Analysis_Type = c("Careful (with Confounder)", "Less Careful (No Confounder)", "Original (Hypergeomet
  Rejection_Rate = c(rate_b, rate_c, rate_d)
)

knitr::kable(results_table, caption = "Comparison of Rejection Rates (Type I Error)")
```

Table 1: Comparison of Rejection Rates (Type I Error)

Analysis_Type	Rejection_Rate
Careful (with Confounder)	0.050
Less Careful (No Confounder)	0.455
Original (Hypergeometric)	0.520